5. Guide for LaTeX Users

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Markups → Functions

- `=` → `#heading`
- `* *` → `#strong`
- `_ _` → `#emph`
- `< >` → `#label`
- `@` → `#ref` or `#cite`
- `` ` ` `` → `#raw`
- `-` → `#list`
- `+` → `#enum`
- `/ <term>:` → `#terms`
- `$ $` → `#equation`

In Typst, function call always follows with parentheses.

Lists

`/ <term>: <description>` – creates an item of **Description** list.

`#enum(numbering: <string>)` – changes the numbering style.

blank line between items – makes spacing of a list wider.

Text formatting

Web-links can be put as they are: https://example.com/.

`#underline[<text>]` – <u>makes the text underlined</u>.

`#lower("TEXT")` – returns lower-case text.

To produce monospaced text, choose a monospaced font or use `$mono(<text>)$`.

To choose a language, use `#text(lang: <string>)`

`#rgb("<code>")` – returns a color, can be used in arguments such as `fill: ...`

`#rect(stroke: red)[Text]` ⟶ Text

`#par(first-line-indent: <length>)` – sets indentation.

`#par(spacing: <length>)` – sets a gap between paragraphs.

`#pad(<arg: value list>)[...]` – takes the same arguments as `margin: (...)` property. Shrinks space if the length's positive and expands if negative. E.g., `#pad(x: -1cm)[<paragraph>]` expands margins in 2cm. The `rest: <length>` argument sets all unsettled sides.

How set rule works. Set rule affects elements if the elements don't have default settings of the property. E.g.:

- `math.equation` doesn't have its default color ⟶ `#set text(fill: red)` will affect `$<equation>$`.
- `math.equation` has its default weight of 450 ⟶ `#show math.equation` is required.

Math

`$cases(a, b, c)$` ⟶ $\begin{cases} a \\ b. \\ c \end{cases}$

`$mat(a, b, c; e,f,g)$` ⟶ $\begin{pmatrix} a & b & c \\ e & f & g \end{pmatrix}$.

`$dots.v$` ⟶ ⋮ `$dots.down$` ⟶ ⋱.

`$arrow.l.squiggly$` == ⇜

To use Typst values in math mode, use `#`, e.g., `$text(fill: #red, x_2)$` ⟶ x_2.

`#set math.frac(style: <str>)` property can be:

- "vertical" → $\frac{a+b}{a-b}$
- "skewed" → $^{a+b}/_{a-b}$
- "horizontal" → $(a+b)/(a-b)$

Escaping is a use of special character to treat the next one differently. In math mode we escape slashes and delimiters using `\`, e.g:

- `$(a+b)\/(a-b)$` → $(a+b)/(a-b)$ in the vertical style.
- `$ \( (a+b)/c ) $` → $(\frac{a+b}{c})$.

`$lr(size: #<n>%, <del><expr><del>)$` controlles the size of delimiters and allows the use of not-parred ones, e.g.:

`$lr(size: #500%, |(x-2)/(x+2)|)$` → $\left|\frac{x-2}{x+2}\right|$

.

`if` and `for`

`#for x in (1, 2, 3) [#x]` → 123.

`#if 0 == 0 [Hello] else [Goodbye]` → Hello.

`#range(4)` → (0, 1, 2, 3) `#range(2, 5)` → (2, 3, 4)

import and include

`#import <str>: <name list>` – import names, use * instead of the list to import all the names. To import from **Typst Universe**, use a string of the following form: `"@preview/<name>:<version>"`, e.g., `#import "@preview/cetz:0.5.2": *` – very powerful drawing tool.

The name conflict solves very simply: the last declaration of the name in current scope is used.

`#include "<file>"` – includes content, not names (variables/ functions).

`#cite(<key>, form: "prose")` – produces a citation that is suitable for inclusion in a sentence.

Questions 1

1. Give corresponding functions for the following markups: =, * *, _ _, < >, @, ` `, -, +, / <term>:, $ $.
2. What does / <term>: <description> markup creates?
3. Explain `enum.numbering` property.
4. What does blank line between list item do to the list?
5. Explain `#underline()` and `#lower()` functions.
6. How to produce monospace text in: content mode, math mode?
7. How to say Typst which language a text is written in?
8. Explain `#rgb()` function.
9. Explain `rect.stroke` property.
10. Explain `par.first-line-indent` property.
11. Explain `par.spacing` property.
12. Explain `#pad()` function.
13. Explain `page.margin.rest` property.
14. Why `#set text(fill: red)` affects math-mode text, but `#set text(weight: "regular")` doesn't? Explain in terms of how set rule works.
15. Explain `cases()` function. Explain `mat()` function.
16. What will be produced by `$dots.v$, $dots.down$, $arrow.l.squiggly$`?
17. Find an error: `$text(fill: red, x_2)`.
18. Explain `math.frac.style` property.
19. What is escaping in the context of programming?
20. How to escape slashes and delimiters in math mode?
21. Explain `lr()` function.
22. Explain `#for` command.
23. Explain `#if` command.

24. Explain `#range()` function.
25. Explain `#import` command.
26. What's * used for in `#import` command and where?
27. How does a string we use with `#import` to import something from Typst Universe look?
28. What's CeTZ package is used for?
29. Explain `#include` command.
30. Explain `#cite()` function.

Questions 2

1. Give markups for the following functions: `#heading()`, `#strong()`, `#emph()`, `#label()`, `#ref()`. `#cite()`, `#raw()`, `#list()`, `#enum()`, `#terms()`, `#equation()`.
2. Which markup created an item of description list?
3. How to make numbered list to be numbered like I, II, III, ...
4. Is `#link()` necessary to put a web-link?
5. How to make text underlined? Which function wraps a string and returns the string with all lowercase?
6. Explain `mono()` function.
7. Explain `text.lang` property.
8. How to produce a color value knowing the RGB code of the color.
9. How to produce Text ?
10. How to make first lines of paragraphs indented?
11. How to set gap between paragraphs?
12. How to make margins of a particular paragraph shrink or expand?
13. What's a selector for math-mode text?
14. What's default weight of math-mode text?
15. How to produce $\begin{cases} a \\ b \\ c \end{cases}$? How to produce $\begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}$?
16. How to produce the symbols: What will be produced by $\vdots$, $\ddots$, $\curvearrowleft$?
17. What's required to use Typst data types such as length, color, alignment, etc., in the math mode?
18. How and what for we use the strings "vertical", "skewed", "horizontal". Hint: it's about math mode.

19. How do we call a use of special character to treat the next one differently?
20. What will be rendered from `$(a+b)\/(a-b)$` and `$ \( (a+b)/c ) $`?
21. How to use not-pared characters such as |...| as scalable delimiters? How to control the scaling?
22. Predict the output: `#for x in (1, 2, 3) [#x]`.
23. Predict the output: `#if 0 == 0 [Hello] else [Goodbye]`.
24. Which function produces array from 0 to `<arg minus 1>`. What does it do when given two arguments?
25. Which command do we use to import names (func/var)?
26. How to import all the manes from the file at once?
27. Explain the string by its parts: `"@preview/cetz:0.5.2"`. Where and what for is it used?
28. How does Typst resolves name conflicts?
29. Which command do we use to import content from another file? Does it imports the names of functions and variables determined in the file?
30. Explain `cite.form` property.